Type MS · Small changes and percentage changes

Answer 1 of 7 · Question 5 part whole · 6 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

5	$\frac{dy}{dx} = e^{-2x} - 2xe^{-2x} = e^{-2x}(1-2x) \Rightarrow \delta y = e^{-2x}(1-2x)\delta x$	M1A1A1
	$\frac{\delta y}{y} \times 100 = \frac{e^{-2x}(1-2x)\delta x}{xe^{-2x}} \times 100 = \frac{e^{-2x}(1-2x) \times 0.03x}{xe^{-2x}} \times 100$	dM1ddM1
	$\%$ change in $y = 3(1-2x)[\%]$	A1
		Total 6 marks

Type MS · Small changes and percentage changes

Answer 1 of 7 · Question 5 part whole · 6 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Small changes and percentage changes

Answer 2 of 7 · Question 5 part whole · 5 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

Number		
5	$A=\pi(3r)^2=9\pi r^2 \Rightarrow \frac{dA}{dr}=18\pi r$	M1
	$\delta A \approx \frac{dA}{dr}\times\delta r = 18\pi r(\delta r)$	B1
	$\frac{\delta A}{A}\approx\frac{18\pi r}{A}\delta r=\frac{18\pi r}{9\pi r^2}\delta r=2\frac{\delta r}{r}$	M1
	So when $\frac{\delta r}{r}=0.05\% \Rightarrow \frac{\delta A}{A}\approx 0.1\%$ so the area increases by about 0.1%	M1A1
	ALT	
	Radius (after increase) $= 3r\times\left(1+\frac{0.05}{100}\right)$	{M1}
	$=3.0015r$	{B1}
	Area before increase $=\pi(3r)^2=9\pi r^2$ Area after increase $=$	
	$A=\pi(3.0015r)^2=9.00900225\pi r^2$	{M1}
	Percentage increase $=\frac{9.00900225\pi r^2-9\pi r^2}{9\pi r^2}\times 100=0.100025\approx 0.1\%$	{M1}
	so the area increases by about 0.1%	{A1}
		Total 5 marks
M1	Differentiate A wrt r	

Type MS · Small changes and percentage changes

Answer 2 of 7 · Question 5 part whole · 5 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Small changes and percentage changes

Answer 3 of 7 · Question 8 part a · 2 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

8 a	$\frac{4}{3}\pi r^3 = 500$ so $r^3 = \frac{3\times 500}{4\pi}$ therefore $r = 4.92$ accept awrt 4.92	M1 A1 (2)
b	$\delta A = 20$	B1
	$(A = 4\pi r^2) \rightarrow \left(\frac{dA}{dr}\right) = 8\pi r$	M1
	$\delta r \approx \frac{dr}{dA}\delta A = 20 \times \frac{1}{8\pi r}$	M1
	When $r = 4.92$ $\delta r \approx \frac{20}{8\pi(4.92)} = 0.16(16204507)$	dM1
	or 0.1617428283 if 4.92 used 0.16 cm accept awrt to 0.16	A1 (5)
		Total 7 marks

Type MS · Small changes and percentage changes

Answer 3 of 7 · Question 8 part a · 2 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Small changes and percentage changes

Answer 4 of 7 · Question 8 part b · 5 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

8 a	$\frac{4}{3}\pi r^3 = 500$ so $r^3 = \frac{3\times 500}{4\pi}$ therefore $r = 4.92$ accept awrt 4.92	M1 A1 (2)
b	$\delta A = 20$	B1
	$(A = 4\pi r^2) \rightarrow \left(\frac{dA}{dr}\right) = 8\pi r$	M1
	$\delta r \approx \frac{dr}{dA}\delta A = 20 \times \frac{1}{8\pi r}$	M1
	When $r = 4.92$ $\delta r \approx \frac{20}{8\pi(4.92)} = 0.16(16204507)$	dM1
	or 0.1617428283 if 4.92 used 0.16 cm accept awrt to 0.16	A1 (5)
		Total 7 marks

Type MS · Small changes and percentage changes

Answer 4 of 7 · Question 8 part b · 5 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Small changes and percentage changes

Answer 5 of 7 · Question 7 part a · 7 marks

2023 JUN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

7(a)	$y=\sqrt{\dfrac{e^{4x}}{2x-3}}=e^{2x}(2x-3)^{-\frac{1}{2}}$	B1
	$y=\dfrac{e^{2x}}{\sqrt{2x-3}} \Rightarrow \dfrac{dy}{dx}=\dfrac{2(2x-3)^{\frac{1}{2}}e^{2x}-2\times\frac{1}{2}\times e^{2x}(2x-3)^{-\frac{1}{2}}}{\left[(2x-3)^{\frac{1}{2}}\right]^2}$	M1A1A1
	$\dfrac{dy}{dx}=\dfrac{\frac{2(2x-3)e^{2x}-2\times\frac{1}{2}\times e^{2x}}{(2x-3)^{\frac{1}{2}}}}{2x-3}=\dfrac{2(2x-3)e^{2x}-2\times\frac{1}{2}\times e^{2x}}{(2x-3)^{\frac{3}{2}}}$	M1 A1
	$=\dfrac{e^{2x}(4x-7)}{(2x-3)^{\frac{3}{2}}}$	A1cso [7]
	$\Rightarrow \delta y \approx \dfrac{e^{2x}(4x-7)}{(2x-3)^{\frac{3}{2}}}\delta x^*$	
(b)	When $x=2.5$, $\delta x=\dfrac{0.2}{100}\times 2.5=0.005$	B1
	$\delta y \approx \dfrac{e^{2\times 2.5}(4\times 2.5-7)}{(2\times 2.5-3)^{\frac{3}{2}}}\times 0.005$	M1
	$\Rightarrow \delta y \approx 0.79$	A1 [3]
		Total 10 marks

Part	Mark	Notes
(a)	**If any candidate attempts part (a) using Chain Rule – please send to Review.**	
	B1	Simplifies the equation into a form which can be differentiated. For example, Award this mark for correct subsequent use in differentiation. $y=\sqrt{\dfrac{e^{4x}}{2x-3}}=e^{2x}(2x-3)^{-\frac{1}{2}}$ or $\dfrac{e^{2x}}{(2x-3)^{\frac{1}{2}}}$ or even $\dfrac{\left(e^{4x}\right)^{\frac{1}{2}}}{(2x-3)^{\frac{1}{2}}}$ or $\dfrac{\left(e^{4x}\right)^{\frac{1}{2}}}{\sqrt{(2x-3)}}$
	Uses Quotient rule. – **NB – This is a 'show' question. Check every line of working.**	
	M1	• The denominator must be correct and squared. • There must be an attempt to differentiate both terms • The two terms in the numerator must be subtracted either way around. Minimally acceptable differentiation is as follows: $e^{2x}\to 2e^{2x}$, $(2x-3)^{\frac{1}{2}}\to k(2x-3)^{-\frac{1}{2}}$ Allow $\left(e^{4x}\right)^{\frac{1}{2}}\to 4e^{4x}\times\frac{1}{2}\times\left(e^{4x}\right)^{-\frac{1}{2}}$

Type MS · Small changes and percentage changes

Answer 5 of 7 · Question 7 part a · 7 marks

2023 JUN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

	A1	One term must be fully correct Either $2(2x-3)^{\frac{1}{2}}e^{2x}$ or $-2\times\frac{1}{2}\times e^{2x}(2x-3)^{-\frac{1}{2}}$
	A1	$\frac{dy}{dx}$ fully correct. Ignore poor notation and erroneous subsequent simplification. $'\frac{dy}{dx}' = \frac{2(2x-3)^{\frac{1}{2}}e^{2x} - 2\times\frac{1}{2}\times e^{2x}(2x-3)^{-\frac{1}{2}}}{2x-3}$ **OR** $'\frac{dy}{dx}' = \frac{(2x-3)^{\frac{1}{2}}\times 4e^{4x}\times\frac{1}{2}\times(e^{4x})^{-\frac{1}{2}} - 2\times\frac{1}{2}\times(e^{4x})^{\frac{1}{2}}(2x-3)^{-\frac{1}{2}}}{2x-3}$
	Uses Product Rule	**NB – This is a 'show' question. Check every line of working.**
	M1	The correct formula must be used. • $\frac{dy}{dx} = uv' + vu'$ • There must be an attempt to differentiate both terms $e^{2x} \to 2e^{2x}$, $(2x-3)^{-\frac{1}{2}} \to k(2x-3)^{-\frac{3}{2}}$ Allow $(e^{4x})^{\frac{1}{2}} \to 4e^{4x}\times\frac{1}{2}\times(e^{4x})^{-\frac{1}{2}}$
	A1	One term must be fully correct Either $(2x-3)^{-\frac{1}{2}}\times 2e^{2x}$ or $\left(-\frac{1}{2}\times 2\times(2x-3)^{-\frac{3}{2}}\right)e^{2x}$
	A1	$\frac{dy}{dx}$ fully correct. Ignore poor notation and erroneous subsequent simplification. $'\frac{dy}{dx}' = (2x-3)^{-\frac{1}{2}}\times 2e^{2x} + \left(-\frac{1}{2}\times 2\times(2x-3)^{-\frac{3}{2}}\right)e^{2x}$ **OR** $'\frac{dy}{dx}' = (2x-3)^{-\frac{1}{2}}\times 4e^{4x}\times\frac{1}{2}\times(e^{4x})^{-\frac{1}{2}} + \left(-\frac{1}{2}\times 2\times(2x-3)^{-\frac{3}{2}}\right)(e^{4x})^{\frac{1}{2}}$
	Simplification – Check their work carefully here.	
	M1	Quotient Rule A correct attempt to simplify the numerator by forming a fraction over $(2x-3)^{\frac{1}{2}}$ Product Rule A correct attempt to simplify by forming a fraction over $(2x-3)^{\frac{3}{2}}$
	A1	For $'\frac{dy}{dx}' = \frac{e^{2x}(4x-7)}{(2x-3)^{\frac{3}{2}}}$
	A1 cso	For the expression exactly as given $\delta y \approx \frac{e^{2x}(4x-7)}{(2x-3)^{\frac{3}{2}}}\delta x$
(b)	**B1**	For finding the change in x
	M1	For using the given expression to substitute the values and evaluate the expression.

Type MS · Small changes and percentage changes

Answer 5 of 7 · Question 7 part a · 7 marks

2023 JUN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

		Do not accept a substitution of 0.2 or 0.2% for δx
	A1	For $'\delta y' \approx 0.79$ accept awrt 0.79 Do not penalise poor notation here. Allow dy = 0.79

	ALT 2	**Uses Chain Rule**
	B1	This mark is scored when they divide through later by e^{2x} and $(2x-3)^{\frac{1}{2}}$
	M1	The correct form must be used. $y = u^{\frac{1}{2}} \quad u' = \frac{4(2x-3)e^{4x} - 2e^{4x}}{(2x-3)^2} = \left[\frac{e^{4x}(8x-14)}{(2x-3)^2}\right] \quad y' = \frac{1}{2}\left(\frac{e^{4x}}{2x-3}\right)^{-\frac{1}{2}}$ $\Rightarrow \frac{dy}{dx} = \frac{1}{2}\left(\frac{e^{4x}}{2x-3}\right)^{-\frac{1}{2}} \times \left(\frac{4(2x-3)e^{4x} - 2e^{4x}}{(2x-3)^2}\right)$ Both terms must be differentiated correctly $e^{4x} \to 4e^{4x}$, $(2x-3) \to k$
	A2	For the correct derivative in unsimplified form. We must see this in full to determine if they simplify further correctly as it is a show question. Please award **both** A marks for the correct derivative seen.
	B1	This mark is awarded at this point. Simplifies $\left(\frac{e^{4x}}{2x-3}\right)^{-\frac{1}{2}}$ to e^{2x} and $(2x-3)^{\frac{1}{2}}$ accept inverted or not.
	M1	Simplifies e^{2x} and $(2x-3)^{\frac{1}{2}}$ to obtain a denominator of $(2x-3)^{\frac{3}{2}}$ $\frac{dy}{dx} = \frac{1}{2}\left(\frac{(2x-3)^{\frac{1}{2}}}{e^{2x}}\right) \times \left(\frac{4(2x-3)e^{4x} - 2e^{4x}}{(2x-3)^2}\right) = \frac{1}{2}\left(\frac{4(2x-3)e^{2x} - 2e^{2x}}{(2x-3)^{\frac{3}{2}}}\right)$
	A1	For the correct expression with no errors.

Type MS · Small changes and percentage changes

Answer 5 of 7 · Question 7 part a · 7 marks

2023 JUN · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Small changes and percentage changes

Answer 6 of 7 · Question 7 part b · 3 marks

2023 JUN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

7(a)	$y=\sqrt{\dfrac{e^{4x}}{2x-3}}=e^{2x}(2x-3)^{-\frac{1}{2}}$	B1
	$y=\dfrac{e^{2x}}{\sqrt{2x-3}} \Rightarrow \dfrac{dy}{dx}=\dfrac{2(2x-3)^{\frac{1}{2}}e^{2x}-2\times\frac{1}{2}\times e^{2x}(2x-3)^{-\frac{1}{2}}}{\left[(2x-3)^{\frac{1}{2}}\right]^2}$	M1A1A1
	$\dfrac{dy}{dx}=\dfrac{\frac{2(2x-3)e^{2x}-2\times\frac{1}{2}\times e^{2x}}{(2x-3)^{\frac{1}{2}}}}{2x-3}=\dfrac{2(2x-3)e^{2x}-2\times\frac{1}{2}\times e^{2x}}{(2x-3)^{\frac{3}{2}}}$	M1 A1
	$=\dfrac{e^{2x}(4x-7)}{(2x-3)^{\frac{3}{2}}}$	A1cso
	$\Rightarrow \delta y \approx \dfrac{e^{2x}(4x-7)}{(2x-3)^{\frac{3}{2}}}\delta x^*$	[7]
(b)	When $x=2.5$, $\delta x=\dfrac{0.2}{100}\times 2.5=0.005$	B1
	$\delta y \approx \dfrac{e^{2\times 2.5}(4\times 2.5-7)}{(2\times 2.5-3)^{\frac{3}{2}}}\times 0.005$	M1
	$\Rightarrow \delta y \approx 0.79$	A1 [3]
		Total 10 marks

Part	Mark	Notes
(a)	**If any candidate attempts part (a) using Chain Rule – please send to Review.**	
	B1	Simplifies the equation into a form which can be differentiated. For example, Award this mark for correct subsequent use in differentiation. $y=\sqrt{\dfrac{e^{4x}}{2x-3}}=e^{2x}(2x-3)^{-\frac{1}{2}}$ or $\dfrac{e^{2x}}{(2x-3)^{\frac{1}{2}}}$ or even $\dfrac{\left(e^{4x}\right)^{\frac{1}{2}}}{(2x-3)^{\frac{1}{2}}}$ or $\dfrac{\left(e^{4x}\right)^{\frac{1}{2}}}{\sqrt{(2x-3)}}$
	Uses Quotient rule. – **NB – This is a 'show' question. Check every line of working.**	
	M1	• The denominator must be correct and squared. • There must be an attempt to differentiate both terms • The two terms in the numerator must be subtracted either way around. Minimally acceptable differentiation is as follows: $e^{2x}\rightarrow 2e^{2x}$, $(2x-3)^{\frac{1}{2}}\rightarrow k(2x-3)^{-\frac{1}{2}}$ Allow $\left(e^{4x}\right)^{\frac{1}{2}}\rightarrow 4e^{4x}\times\frac{1}{2}\times\left(e^{4x}\right)^{-\frac{1}{2}}$

Type MS · Small changes and percentage changes

Answer 6 of 7 · Question 7 part b · 3 marks

2023 JUN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

	A1	One term must be fully correct Either $2(2x-3)^{\frac{1}{2}}e^{2x}$ or $-2\times\frac{1}{2}\times e^{2x}(2x-3)^{-\frac{1}{2}}$
	A1	$\frac{dy}{dx}$ fully correct. Ignore poor notation and erroneous subsequent simplification. $'\frac{dy}{dx}' = \frac{2(2x-3)^{\frac{1}{2}}e^{2x} - 2\times\frac{1}{2}\times e^{2x}(2x-3)^{-\frac{1}{2}}}{2x-3}$ **OR** $'\frac{dy}{dx}' = \frac{(2x-3)^{\frac{1}{2}}\times 4e^{4x}\times\frac{1}{2}\times(e^{4x})^{-\frac{1}{2}} - 2\times\frac{1}{2}\times(e^{4x})^{\frac{1}{2}}(2x-3)^{-\frac{1}{2}}}{2x-3}$
	Uses Product Rule	**NB – This is a 'show' question. Check every line of working.**
	M1	The correct formula must be used. • $\frac{dy}{dx} = uv' + vu'$ • There must be an attempt to differentiate both terms $e^{2x} \to 2e^{2x}$, $(2x-3)^{-\frac{1}{2}} \to k(2x-3)^{-\frac{3}{2}}$ Allow $(e^{4x})^{\frac{1}{2}} \to 4e^{4x}\times\frac{1}{2}\times(e^{4x})^{-\frac{1}{2}}$
	A1	One term must be fully correct Either $(2x-3)^{-\frac{1}{2}}\times 2e^{2x}$ or $\left(-\frac{1}{2}\times 2\times(2x-3)^{-\frac{3}{2}}\right)e^{2x}$
	A1	$\frac{dy}{dx}$ fully correct. Ignore poor notation and erroneous subsequent simplification. $'\frac{dy}{dx}' = (2x-3)^{-\frac{1}{2}}\times 2e^{2x} + \left(-\frac{1}{2}\times 2\times(2x-3)^{-\frac{3}{2}}\right)e^{2x}$ **OR** $'\frac{dy}{dx}' = (2x-3)^{-\frac{1}{2}}\times 4e^{4x}\times\frac{1}{2}\times(e^{4x})^{-\frac{1}{2}} + \left(-\frac{1}{2}\times 2\times(2x-3)^{-\frac{3}{2}}\right)(e^{4x})^{\frac{1}{2}}$
	Simplification – Check their work carefully here.	
	M1	Quotient Rule A correct attempt to simplify the numerator by forming a fraction over $(2x-3)^{\frac{1}{2}}$ Product Rule A correct attempt to simplify by forming a fraction over $(2x-3)^{\frac{3}{2}}$
	A1	For $'\frac{dy}{dx}' = \frac{e^{2x}(4x-7)}{(2x-3)^{\frac{3}{2}}}$
	A1 cso	For the expression exactly as given $\delta y \approx \frac{e^{2x}(4x-7)}{(2x-3)^{\frac{3}{2}}}\delta x$
(b)	**B1**	For finding the change in x
	M1	For using the given expression to substitute the values and evaluate the expression.

Type MS · Small changes and percentage changes

Answer 6 of 7 · Question 7 part b · 3 marks

2023 JUN · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

		Do not accept a substitution of 0.2 or 0.2% for δx
	A1	For $'\delta y' \approx 0.79$ accept awrt 0.79 Do not penalise poor notation here. Allow dy = 0.79

	ALT 2 Uses Chain Rule	
	B1	This mark is scored when they divide through later by e^{2x} and $(2x-3)^{\frac{1}{2}}$
	M1	The correct form must be used. $y = u^{\frac{1}{2}} \quad u' = \frac{4(2x-3)e^{4x} - 2e^{4x}}{(2x-3)^2} = \left[\frac{e^{4x}(8x-14)}{(2x-3)^2}\right] \quad y' = \frac{1}{2}\left(\frac{e^{4x}}{2x-3}\right)^{-\frac{1}{2}}$ $\Rightarrow \frac{dy}{dx} = \frac{1}{2}\left(\frac{e^{4x}}{2x-3}\right)^{-\frac{1}{2}} \times \left(\frac{4(2x-3)e^{4x} - 2e^{4x}}{(2x-3)^2}\right)$ Both terms must be differentiated correctly $e^{4x} \to 4e^{4x}$, $(2x-3) \to k$
	A2	For the correct derivative in unsimplified form. We must see this in full to determine if they simplify further correctly as it is a show question. Please award **both** A marks for the correct derivative seen.
	B1	This mark is awarded at this point. Simplifies $\left(\frac{e^{4x}}{2x-3}\right)^{-\frac{1}{2}}$ to e^{2x} and $(2x-3)^{\frac{1}{2}}$ accept inverted or not.
	M1	Simplifies e^{2x} and $(2x-3)^{\frac{1}{2}}$ to obtain a denominator of $(2x-3)^{\frac{3}{2}}$ $\frac{dy}{dx} = \frac{1}{2}\left(\frac{(2x-3)^{\frac{1}{2}}}{e^{2x}}\right) \times \left(\frac{4(2x-3)e^{4x} - 2e^{4x}}{(2x-3)^2}\right) = \frac{1}{2}\left(\frac{4(2x-3)e^{2x} - 2e^{2x}}{(2x-3)^{\frac{3}{2}}}\right)$
	A1	For the correct expression with no errors.

Type MS · Small changes and percentage changes

Answer 6 of 7 · Question 7 part b · 3 marks

2023 JUN · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Small changes and percentage changes

Answer 7 of 7 · Question 8 part b · 3 marks

2023 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

8(a)	$\left(\frac{dy}{dx}=\right)\frac{5x^2\times2\times3\times e^{3x+1}-10x\times2e^{3x+1}}{(5x^2)^2}$ oe	M1A1A1
	$\left(\frac{dy}{dx}=\right)\frac{xe^{3x+1}(30x-20)}{25x^4}$ or $\frac{10xe^{3x+1}(3x-2)}{25x^4}$	dM1
	$=\left[\frac{10e^{3x+1}(3x-2)}{25x^3}\right]=\frac{2e^{3x+1}(3x-2)}{5x^3}$	A1 [5]
ALT – use of product rule	$\left(\frac{dy}{dx}=\right)2e^{3x+1}(-1)\times2\times5x\times(5x^2)^{-2}+2\times3e^{3x+1}(5x^2)^{-1}$ oe	M1A1A1
	$=x\times(5x^2)^{-2}e^{3x+1}(-20+30x)$ or $10x\times(5x^2)^{-2}e^{3x+1}(-2+3x)$	dM1
	$=\left[\frac{10e^{3x+1}(3x-2)}{25x^3}\right]=\frac{2e^{3x+1}(3x-2)}{5x^3}$	A1 [5]
(b)	$\left(\frac{\delta y}{y}\times100=\right)"\frac{2xe^{3x+1}(3x-2)}{5x^4}"\times\delta x\times\frac{100}{y}$ or $"\frac{2e^{3x+1}(3x-2)}{5x^3}"\times\delta x\times\frac{100}{y}$ oe	M1
	or $(\delta y=)"\frac{2xe^{3x+1}(3x-2)}{5x^4}"\times\delta x$ or $"\frac{2e^{3x+1}(3x-2)}{5x^3}"\times\delta x$ or $y\times\frac{(3x-2)}{5x}\times\delta x$ δx may also be written as $0.02x$	
	$\left(\frac{\delta y}{y}\times100=\right)\frac{2e^{3x+1}(3x-2)}{5x^3}\times0.02x\times\frac{100}{\frac{2e^{3x+1}}{5x^2}}$ oe	dM1
	or $y\times\frac{(3x-2)}{x}\times0.02x\times\frac{100}{y}$ $(\%\text{ change in } y=)6x-4$	A1 [3]
		Total 8 marks

Type MS · Small changes and percentage changes

Answer 7 of 7 · Question 8 part b · 3 marks

2023 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For attempting to use the quotient rule. The definition of an attempt is as follows: • The denominator must be correct. • The terms in the numerator subtracted either way around The expression **must** be of the form: $\frac{K\times x^2\times e^{3x+1}-L\times x\times e^{3x+1}}{\left(5x^2\right)^2}$ $K\neq 10$ or 15 and $L\neq 10$ and $K,L>0$ Note K and L may not be simplified and will often need to be checked.
	A1	Following M1 (general principle of marking), at least one fully correct simplified or unsimplified term on the numerator.
	A1	Fully correct **unsimplified** derivative
	dM1	For correctly taking out common factors of x and e^{3x+1} and attempting to simplify. A numerical factor does not need to be taken out at this point. Dependent on the 1st method mark.
	A1	For the correct derivative in the required form.
ALT use of product rule	M1	For attempting to use the product rule to give an expression of the form $Me^{3x+1}x\times(5x^2)^{-2}+Ne^{3x+1}(5x^2)^{-1}$ $M<0$ and $\neq 4,10$ $N>0$ and $\neq 2,3$
	A1	Following M1 (general principle of marking), at least one term is fully correct.
	A1	Fully correct **unsimplified** derivative
	M1	For correctly taking out common factors of x and $(5x^2)^{-2}$ and e^{3x+1} and attempting to simplify. A numerical factor does not need to be taken out at this point. Dependent on the 1st method mark.
	A1	For the correct derivative in the required form.
For the final M mark in particular – any work felt to be of credit not fitting the mark scheme – send to Review.		
(b)	M1	For any of the statements shown, using their answer from part (a), which must be of the form: $\frac{Ae^{3x+1}(Bx-C)}{Dx^3}$ Note, we'll allow candidates to use an expression here that is not quite of the form given in part (a). This mark may be implied by further correct working.
	dM1	For substituting in the expression for y **and** a correct attempt to simplify their expression to the required form. You must check their working here, do not allow an answer that has come from incorrect working (general principle anyway, but extra care needed for this question). Condone 100 written as 100% if the candidate is correctly multiplying by 100. Dependent on previous method mark.
	A1	For the correct expression as shown.

Type MS · Small changes and percentage changes

Answer 7 of 7 · Question 8 part b · 3 marks

2023 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks